Journal of Science and Medicine in Sport 27 (2024) 187–196

Contents lists available at ScienceDirect

Journal of Science and Medicine in Sport

journal homepage: www.elsevier.com/locate/jsams

Review

Comparative efficacy of physical activity types on executive functions in children and adolescents: A network meta-analysis of randomized controlled trials

Junyu Wang [a,b,1], Yong Yang [c], Longkai Li [d], Xuemin Yang [b], Xuejun Guo [b], Xun Yuan [b], Ting Xie [e], Kunyi Yang [a], Jie Zhuang [a,*]

[a] School of Exercise and Health, Shanghai University of Sport, China
[b] Officers College of PAP, China
[c] Institute for Brain Sciences Research, School of Life Sciences, Henan University, China
[d] Putuo District Sports Administration Center of Shanghai, China
[e] Chengdu Xinqiao Primary School, China

ARTICLE INFO

Article history:
Received 24 June 2023
Received in revised form 4 November 2023
Accepted 13 November 2023
Available online 22 November 2023

Keywords:
Child
Adolescent
Physical activity
Cognition
Memory
Inhibition

ABSTRACT

Objectives: This study investigates the impact of various physical activity (PA) types on executive functions (EFs) in children and adolescents.
Design: A systematic review and network meta-analysis of randomized clinical trials.
Methods: We searched databases such as PubMed, Embase, Cochrane, and Web of Science up to April 2023, including randomized controlled trials involving 6 distinct PA types for healthy children and adolescents. The Cochrane risk of bias tool was used to assess the risk of bias, and a random-effects model in STATA 17.0 was used to calculate standardized mean differences (SMDs) and 95 % confidence intervals (CI).
Results: Ball Games emerged as the most effective modality for improving updating accuracy, securing a SUCRA score of 94.4 %, and for reducing inhibition reaction time, with a SUCRA score of 94.8 %. Cognitively Engaging Physical Activity led in improving inhibition accuracy with a SUCRA score of 71.7 %. Dance excelled in improving update accuracy and reducing shifting reaction time, with SUCRA scores of 86.6 % and 99.5 %, respectively.
Conclusions: PA has a significant benefit in EFs in children and adolescents, however the size of the effect varies by type of PA. Ball Games emerged as the most efficacious modality for enhancing updating accuracy and for expediting inhibition reaction time. Cognitively Engaging Physical Activity proved to be the preeminent strategy for improving inhibition accuracy. Dance was distinguished as the optimal approach for improving updating accuracy and reducing shifting reaction time.
© 2023 The Author(s). Published by Elsevier Ltd on behalf of Sports Medicine Australia. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

Practical implications

1. Targeting updating and inhibition reaction time with Ball Games: Given our NMA findings that Ball Games excel in improving both updating accuracy and inhibition reaction time, educational institutions and caregivers should give consideration to incorporating Ball Games into daily routines or curricula. This could significantly enhance working memory and inhibition speed, both crucial for academic performance and cognitive development in children and adolescents.
2. Improving inhibition accuracy with Cognitively Engaging Physical Activity: Schools and youth centers should view Cognitively Engaging Physical Activity as a valuable addition to their physical and cognitive training programs. Our results indicate that Cognitively Engaging Physical Activity is especially effective in enhancing inhibition accuracy, thereby reinforcing its role as an essential tool for boosting cognitive control in young individuals.
3. Enhancing updating accuracy and shifting reaction time with Dance: The high efficacy of Dance in improving updating accuracy and shifting reaction time underscores the need for its inclusion in youth training programs. DC activities could range from complex team sports to tasks involving multi-step problem-solving, offering a versatile approach to enhancing working memory and cognitive flexibility.

1. Introduction

Executive functions (EFs) are crucial cognitive abilities that manage and control complex, goal-oriented thought processes and actions. They

* Corresponding author.
E-mail address: zhuangjiesh@163.com (J. Zhuang).
[1] The authors contributed equally to this study.

https://doi.org/10.1016/j.jsams.2023.11.006
1440-2440/© 2023 The Author(s). Published by Elsevier Ltd on behalf of Sports Medicine Australia. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

include abilities such as updating, inhibiting, and shifting.[1] They play a pivotal role in the growth and development of children and adolescents, enabling them to concentrate, multitask, comprehend and manage their emotions, and bolster their social interaction skills.[2,3] However, contemporary challenges such as increased academic pressure, augmented screen time, and diminished physical activity (PA) are posing adverse effects on the development of Executive Functions (EFs) in children,[4–6] these negative implications are associated with a range of behavioral and emotional problems, including Attention Deficit Hyperactivity Disorder (ADHD) and aggressive behavior,[7] which often places great physical and emotional strain on the children's families. Therefore, it is crucial to find effective ways to enhance EFs in children and adolescents.

Psychostimulant medications, such as methylphenidate, are clinically effective for treating impaired EFs,[8,9] yet they often come with substantial costs and potential side effects.[10–13] These drawbacks can hinder healthy growth and the establishment of positive health behaviors in children. PA, as an evidence-based alternative, has been substantiated to have beneficial effects on EFs in children.[14] Chronic interventions involving PA—defined as programs extending beyond six weeks and conducted multiple times a week—have been shown to contribute to the holistic health development of children and adolescents.[15–19] However, consensus is lacking on the most effective type of PA for this purpose. Evidence frequently points to the beneficial impact of aerobic exercises on EFs[20] and some studies have favored mixed-motor exercise that includes balance.[21] In recent studies, it has been demonstrated that physical activities incorporating cognitive challenges, such as calculation or memory tasks performed concurrently with PA, have shown remarkable efficacy in enhancing executive functions in adolescents.[22,23]

To address the existing gaps in our understanding of the relative impact of different types of PA on children's EFs a more comprehensive synthesis of evidence is required. Traditional pairwise meta-analyses are limited to comparisons between two distinct intervention approaches, potentially excluding numerous randomized controlled trials (RCTs) that meet inclusion criteria. Network meta-analysis (NMA) offers a robust solution to this challenge, enabling multiple pairwise comparisons within a single statistical model. This approach facilitates a more comprehensive and nuanced evaluation of intervention strategies[24] and is considered the highest level of evidence in treatment guidelines.[25] Although there are existing NMAs that investigate the impact of PA modalities on EFs, these are confined to populations either over 18 years of age[26] or those with ADHD.[27] Therefore, there remains a notable gap in the literature regarding the effects on typically developing children and adolescents.

In light of this, we conducted a systematic review and network meta-analysis to assess the impact of different PAs on EFs (updating, inhibition, and shifting) in children and adolescents. Our aim was to identify the optimal PA regimen for this population.

2. Methods

Our systematic review and network meta-analysis were performed in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) guidelines.[28] And, a pre-specified protocol was followed for this study (PROSPERO registration number: CRD42023432224).

2.1. Search strategy

We searched PubMed, Cochrane, Embase and Web of Science (WOS) databases from inception until April 2023 using relevant search terms, including 'exercise', 'physical activity', 'executive function', 'child', 'adolescent' and 'randomized controlled trial'. The reference lists of identified studies were also examined to find additional pertinent studies. The specific search strategies, including search terms, dates, and processes, are presented in Supplementary A.

2.2. Selection criteria

To be eligible for inclusion in this systematic review and network meta-analysis, studies had to satisfy the following predetermined criteria: (1) Participants had to be healthy children aged between 6 and 12 years or adolescents aged 13–18 years; (2) The PA intervention evaluated in the study must have been a primary focus of the research, and the type of PA must align with one of the six pre-determined categories established by the investigative team. These categories are detailed in Supplementary B; (3) The control group needed to be engaged in a standardized program involving regular PA sessions and non-motor cognitive education; (4) The intervention period was mandated to exceed a duration of six weeks[29]; (5) The primary outcome measured had to be performance in EFs, which could encompass updating, inhibition, shifting; (6) The study needed to be a published RCTs; and (7) The manuscript had to be written in English and published in a peer-reviewed journal.

Conversely, studies were excluded from consideration if they met any of the following conditions: (1) The participant population was diagnosed with obesity, depression, or ADHD, as these disorders may have biased the results; (2) The intervention was predicated on an acute PA regimen (e.g., single-session PA) or a multicomponent program that integrated diet, sleep, or other controls extraneous to PA; (3) The study results did not permit the extraction or estimation of statistical measures such as mean and standard deviation (SD); and (4) There was an inability to access the full text or directly correspond with the authors for additional information.

Title/abstract and full-text screening were conducted independently and in duplicate by investigators (J.Y.W and X.M.Y), with disagreements resolved by discussion or adjudication by a third author (Y.Y).

2.3. Data extraction and bias assessment

Two independent researchers (J.Y.W. and L.K.L.) extracted pertinent data and populated it into an Excel spreadsheet. This extraction encompassed various attributes such as the number of participants, their ages, the nature of interventions undertaken, and both pre- and post-intervention mean scores and standard deviations (SDs) for tests related to EFs. Specifically, EF metrics were grouped into three core subdomains: updating, inhibition and shifting. Updating, also known as working memory, plays a central role in the short-term acquisition, retention and manipulation of information in children and adolescents.[26] Inhibition facilitates selective attention and cognitive control,[2,30] while shifting, also known as cognitive flexibility, enables transitions between different mental operations and thought patterns.[31]

For studies that only reported means and SDs graphically, data extraction was performed using Web Plot Digitizer version 4 (Free Software Foundation, Boston, MA, USA).[32] Where standard errors were reported, these were converted to standard deviations for consistency. Effect sizes for each included study were calculated based on the mean and scaled values of the differences between pre- and post-intervention scores in both the intervention and control groups. The formulas used to calculate these mean change and SD values were: 'Meanchange = Meanpost − Meanpre' and 'SDchange = SQRT [(SDpre2 + SDpost2) − (2 × Corr × SDpre × SDpost)]', with the correlation coefficient (Corr) set at 0.5 according to the Cochrane Handbook guidelines.[33]

Where studies used multiple tasks to assess a single domain of EFs, the analysis prioritized the effect size from the most commonly used task. Furthermore, when a study produced multiple outcome measures from a single task, the measure that required the greatest cognitive load, such as incongruent trials in the Flanker task, was selected.[26,34] This approach was designed to maintain methodological consistency and increase the rigor of EFs assessment across studies.

Three researchers (X.J.G, T.X and X.Y) assessed and rated studies using the Cochrane Risk of Bias Tool, with each trial rated as having low,

unclear, or high risk of bias in each domain. Any disagreements were resolved through mutual agreement.

2.4. Statistical analysis

To account for the variability in scoring scales and measurement methods employed across the included studies, we used standardized mean differences (SMDs) with a 95 % confidence interval as the effect size for comparing EFs across different types of PA. SMDs serve as a robust metric for quantifying the differences in outcomes between PA modalities, thereby providing direct insight into the impact of the intervention on EFs. This approach ensures a methodologically rigorous and comparable assessment of EFs across studies.

The frequentist random-effects NMA model was utilized, and the restricted maximum likelihood estimation method was used. The network suite of commands within the Stata software (version 17, StataCorp, TX, USA) was applied to perform NMA.

Pairwise meta-analyses were conducted using the DerSimonian and Laird random-effects model[35] to compare the intervention effects across various PA types. We assessed inconsistencies between direct and indirect evidence using loop-specific inconsistency, side-splitting, and design-by-treatment interaction models. To evaluate the relative effectiveness of each intervention, the surface under the cumulative ranking curve (SUCRA) was calculated. Additionally, Egger's test was performed to detect any publication bias by initially ranking interventions based on their effectiveness.

3. Results

3.1. Selection, inclusion, and characteristics of studies

From an initial pool of 6220 articles, 31 RCTs were selected for our study based on the inclusion criteria (Fig. 1). Our study analyzed 23,569 participants with an average age of 10.14 years (SD 2.46), the duration of the PA period ranged from 6 to 40 weeks (mean period 16.58 weeks, SD 10.42). PA intervention frequency varied from 1 to 10 times per week (mean frequency 3.32 times/week, SD 1.71), while the duration of a single session spanned between 10 and 150 min (mean time 47.90 min, SD 32.12). Results pertaining to the accuracy of the updating component were available in 21 studies. Meanwhile, data on both accuracy and reaction time for the inhibition component were reported in 20 and 16 studies, respectively. For the shifting component, accuracy and reaction time were documented in 9 and 6 studies, respectively. However, due to the limited number of articles addressing reaction time in updating, we could not include them in the statistical analysis. Detailed characteristics of the included studies are presented in Supplementary C.

3.2. Updating

We conducted a comprehensive meta-analysis involving 21 studies and 21,353 participants to assess the impact of various PA types on the accuracy of updating. The network plots delineated in Fig. 2a, encompass 7 distinct nodes and feature 8 head-to-head comparisons. These studies focused on AE and CEPA as the primary PA interventions.

Table 1a reveals that BG (Ball Game) were significantly more effective than ME (Mixed Exercise) (SMDs = 1.01, 95 % CI [0.04, 1.98]), AE (Aerobic Exercise) (SMDs = 1.32, 95 % CI [0.51, 2.13]), CEPA (Cognitively Engaging Physical Activity) (SMDs = 1.47, 95 % CI [0.67, 2.27]), HIIT (High-Intensity Interval Training) (SMDs = 1.53, 95 % CI [0.50, 2.57]), and CON (Control Group) (SMDs = 1.57, 95 % CI [0.85, 2.30]) in enhancing updating accuracy. Similarly, DC (Dance) outperformed AE (SMDs = 1.03, 95 % CI [0.14, 1.91]), CEPA (SMDs = 1.18, 95 % CI [0.31, 2.05]), HIIT (SMDs = 1.24, 95 % CI [0.15, 2.33]), and CON (SMDs = 1.28, 95 % CI [0.47, 2.09]) in the same metric.

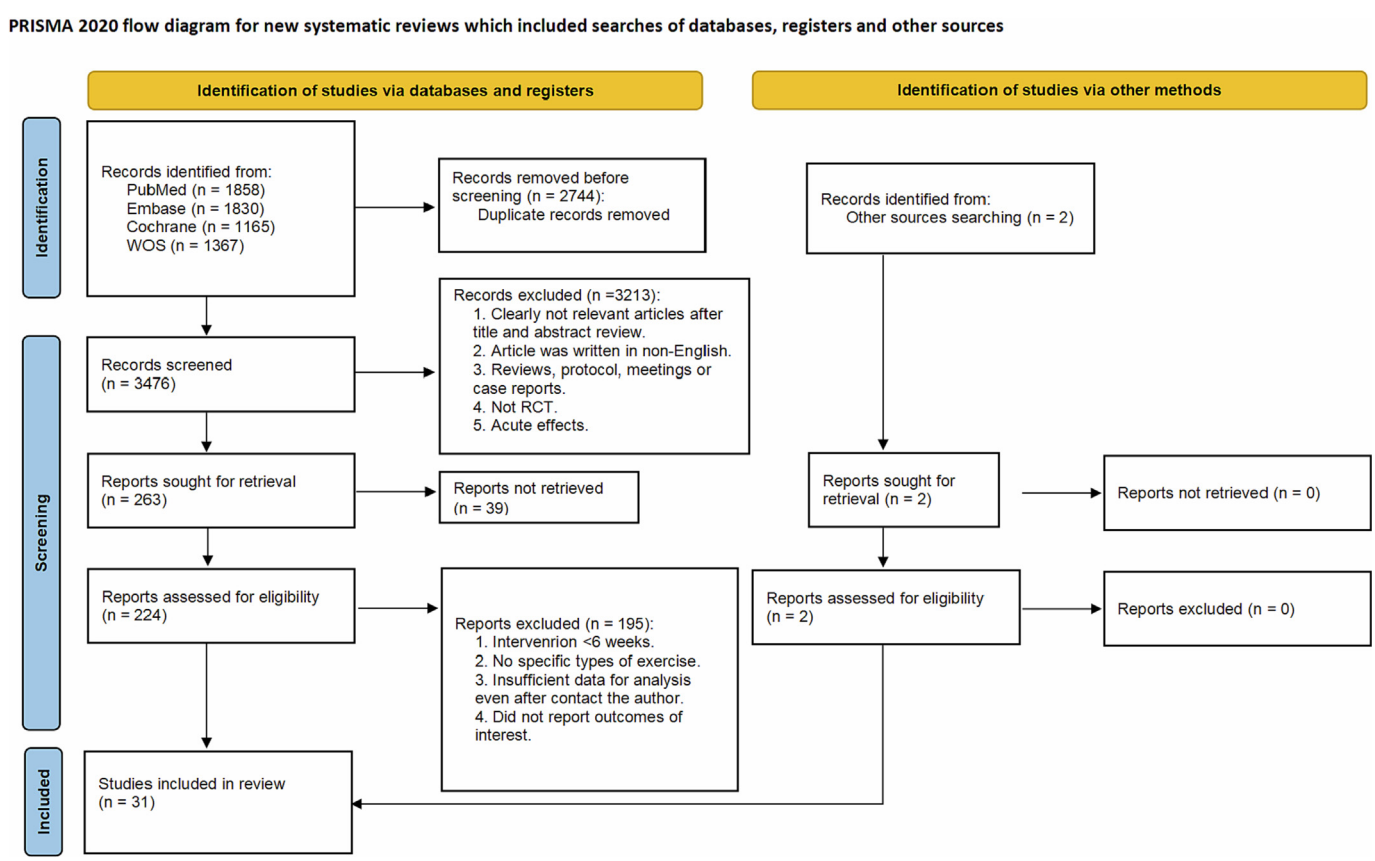

Fig. 1. PRISMA Flow diagram of the search process for studies.

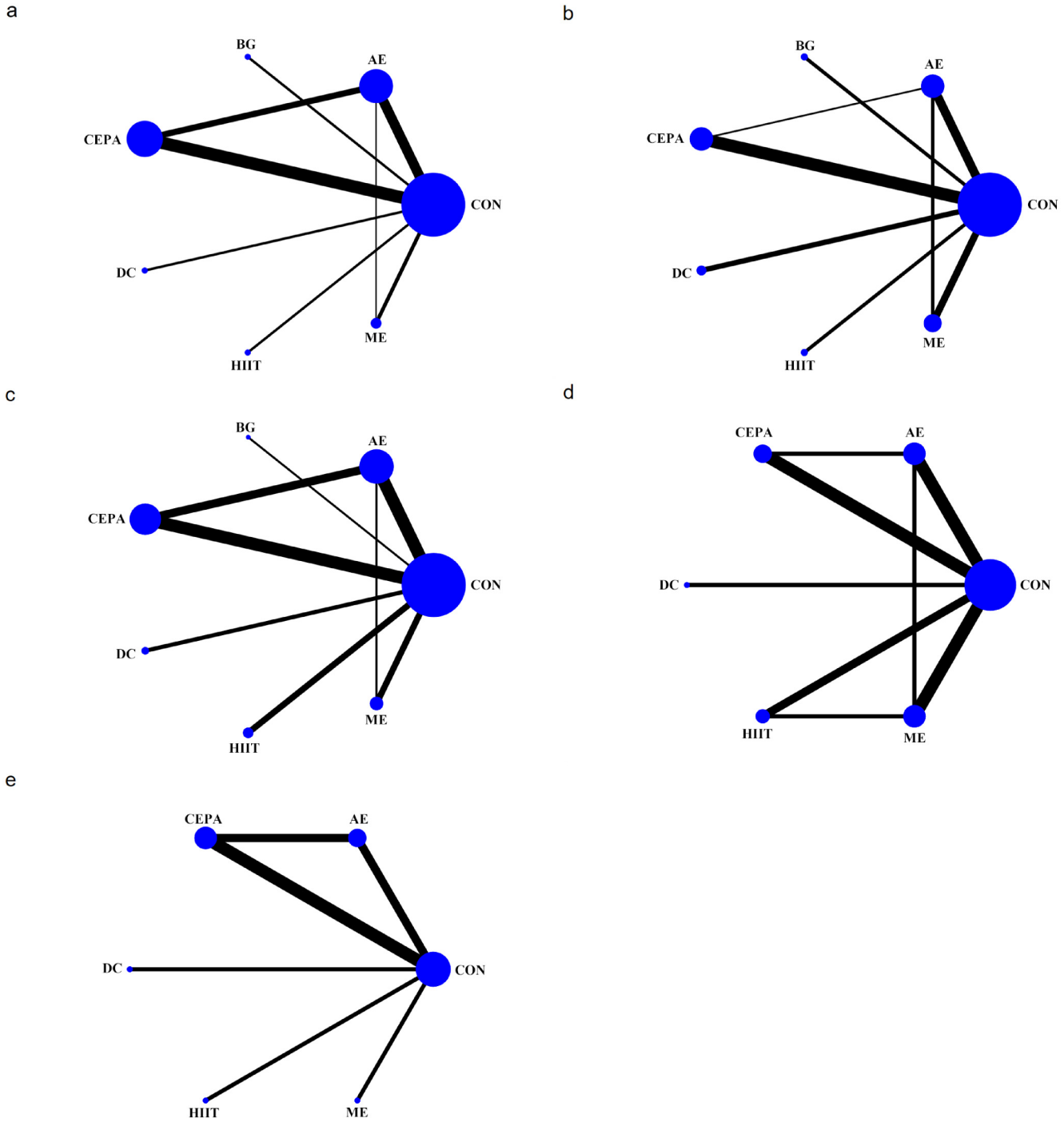

Fig. 2. Network of eligible treatment comparisons for a: updating accuracy, b: inhibition accuracy, c: inhibition reaction time, d: shifting accuracy, e: shifting reaction time.

Subsequent SUCRA probability analysis, as illustrated in Fig. 3a, indicated that BG, with a SUCRA score of 94.4 %, were the most efficacious in improving updating accuracy. They were followed, in descending order of effectiveness, by DC, ME, AE, CEPA, HIIT, and CON.

3.3. Inhibition

In 25 studies involving 21,519 participants, the effects of accuracy or reaction time on inhibition outcomes were evaluated. The network plots, illustrated in Fig. 2b and c, consist of 7 distinct nodes representing various PA types and include 8 direct, head-to-head comparisons for both accuracy and reaction time. Among the array of studies, AE and CEPA emerged as the most frequently compared PA types.

Table 1b and c reveals that CEPA significantly improved inhibition accuracy relative to the CON (SMDs = 0.39, 95 % CI [0.03, 0.75]). However, no significant differences in response times were observed to be evident across PA types in pairwise comparisons.

Subsequent SUCRA probability analysis indicated that the most efficacious PA type for enhancing inhibition accuracy was CEPA (SUCRA score of 71.7 %), followed closely by ME (SUCRA score of 70.5 %) and DC (SUCRA score of 62.9 %). For the reaction time reduction, BG emerged as the most effective (SUCRA score of 94.8 %), followed by CEPA (SUCRA score of 70.5 %) and HIIT (SUCRA score of 52.6 %). Conversely, ME was identified as the least effective approach for improving inhibition reaction time, with a SUCRA score of 15.1 %.

Table 1
League table of executive functions.

a updating accuracy

BG	—	—	—	—	—	—
0.29 (-0.80,1.38)	DC	—	—	—	—	—
1.01 (0.04,1.98)	0.72 (-0.32,1.75)	ME	—	—	—	—
1.32 (0.51,2.13)	**1.03 (0.14,1.91)**	0.31 (-0.38,1.01)	AE	—	—	—
1.47 (0.67,2.27)	**1.18 (0.31,2.05)**	0.46 (-0.24,1.17)	0.15 (-0.25,0.55)	CEPA	—	—
1.53 (0.50,2.57)	**1.24 (0.15,2.33)**	0.53 (-0.45,1.50)	0.21 (-0.61,1.03)	0.07 (-0.74,0.87)	HIIT	—
1.57 (0.85,2.30)	**1.28 (0.47,2.09)**	0.57 (-0.07,1.21)	0.25 (-0.11,0.62)	0.10 (-0.22,0.43)	0.04 (-0.70,0.78)	CON

b inhibition accuracy

CEPA	—	—	—	—	—	—
0.01 (-0.60,0.62)	ME	—	—	—	—	—
0.07 (-0.46,0.61)	0.07 (-0.52,0.65)	AE	—	—	—	—
0.15 (-0.55,0.84)	0.14 (-0.63,0.91)	0.07 (-0.66,0.81)	DC	—	—	—
0.30 (-0.55,1.16)	0.30 (-0.62,1.22)	0.23 (-0.65,1.12)	0.16 (-0.81,1.13)	BG	—	—
0.37 (-0.39,1.13)	0.36 (-0.47,1.20)	0.30 (-0.50,1.10)	0.22 (-0.67,1.12)	0.06 (-0.96,1.09)	HIIT	—
0.39 (0.03,0.75)	0.38 (-0.12,0.89)	0.32 (-0.12,0.75)	0.24 (-0.35,0.84)	0.09 (-0.69,0.86)	0.02 (-0.65,0.69)	CON

c inhibition reaction time

BG	—	—	—	—	—	—
-0.49 (-1.21,0.23)	CEPA	—	—	—	—	—
-0.56 (-1.26,0.14)	-0.07 (-0.33,0.18)	HIIT	—	—	—	—
-0.58 (-1.30,0.14)	-0.09 (-0.37,0.18)	-0.02 (-0.25,0.22)	DC	—	—	—
-0.60 (-1.31,0.11)	-0.11 (-0.28,0.07)	-0.03 (-0.24,0.18)	-0.02 (-0.27,0.24)	AE	—	—
-0.61 (-1.30,0.08)	-0.12 (-0.32,0.09)	-0.04 (-0.15,0.06)	-0.03 (-0.23,0.18)	-0.01 (-0.17,0.15)	CON	—
-0.71 (-1.44,0.03)	-0.22 (-0.52,0.09)	-0.14 (-0.42,0.13)	-0.13 (-0.44,0.19)	-0.11 (-0.39,0.17)	-0.10 (-0.35,0.15)	BG

d shifting accuracy

AE	—	—	—	—	—
-0.16 (-2.26,1.95)	DC	—	—	—	—
0.48 (-0.77,1.72)	0.63 (-1.47,2.74)	ME	—	—	—
0.64 (-0.62,1.90)	0.80 (-1.32,2.92)	0.16 (-1.22,1.55)	CEPA	—	—
0.79 (-0.74,2.31)	0.94 (-1.28,3.17)	0.31 (-1.07,1.69)	0.14 (-1.43,1.71)	HIIT	—
0.90 (-0.08,1.88)	1.05 (-0.81,2.92)	0.42 (-0.56,1.40)	0.25 (-0.76,1.26)	0.11 (-1.10,1.32)	CON

e shifting reaction time

DC	—	—	—	—	—
-0.76 (-1.44,-0.09)	CEPA	—	—	—	—
-0.90 (-1.56,-0.25)	-0.14 (-0.30,0.02)	HIIT	—	—	—
-0.97 (-1.67,-0.27)	-0.20 (-0.45,0.04)	-0.07 (-0.31,0.18)	AE	—	—
-0.97 (-1.62,-0.32)	**-0.21 (-0.36,-0.05)**	**-0.07 (-0.10,-0.04)**	-0.00 (-0.25,0.24)	CON	—
-0.99 (-1.81,-0.35)	**-0.44 (-0.75,-0.13)**	**-0.30 (-0.57,-0.03)**	-0.24 (-0.60,0.13)	-0.23 (-0.50,0.03)	ME

Note: a updating accuracy, b inhibition accuracy, c inhibition reaction time, d shifting accuracy and e shifting reaction time. Significant results are shown in boldface type. AE, Aerobic Exercise; BG, Ball Games; CEPA, Cognitively Engaging Physical Activity; CON, Control Group; DC, Dance; HIIT, High-Intensity Interval Training; ME, Mixed Exercise.

Fig. 3. Ranking of PA intervention strategies based on probability of their effects for a: updating accuracy, b: inhibition accuracy, c: inhibition reaction time, d: shifting accuracy, e: shifting reaction time.

3.4. Shifting

In 9 studies involving 16,876 participants, we scrutinized data to assess the efficacy of various types of PA on shifting outcomes. The network diagram, delineated in Fig. 2d and e, features 6 distinct nodes representing PA types, along with 8 and 6 direct head-to-head comparisons for accuracy and reaction time, respectively. Among these, CEPA was the most recurrently assessed PA type.

Our analyses revealed no significant disparities in shifting accuracy across different PA types. Regarding reaction time in shifting tasks, DC exhibited a significant improvement over CEPA (SMDs = −0.76, 95 % CI [−1.44, −0.09]), HIIT (SMDs = −0.90, 95 % CI [−1.56, −0.25]), AE (SMDs = −0.97, 95 % CI [−1.67, −0.27]), CON (SMDs = −0.97, 95 % CI [−1.62, −0.32]), and ME (SMDs = −0.99, 95 % CI [−1.81, −0.35]). Moreover, CEPA outperformed CON (SMDs = −0.21, 95 % CI [−0.36, −0.05]) and ME (SMDs = −0.44, 95 % CI [−0.75, −0.13]) in

reducing reaction time. HIIT also demonstrated a significant reduction in reaction time when compared to CON (SMDs = −0.07, 95 % CI [−0.10, −0.04]) and ME (SMDs = −0.30, 95 % CI [−0.57, −0.03]).

In terms of enhancing shifting accuracy, AE emerged as the most effective modality (Fig. 3d, SUCRA 77.5 %), followed by DC (SUCRA 74.8 %), ME (SUCRA 50.1 %), CEPA (SUCRA 41.5 %), HIIT (SUCRA 33.9 %), and CON (SUCRA 22.2 %). For the reaction time reduction, DC ranked the highest (Fig. 3e, SUCRA 99.5 %), succeeded by CEPA (SUCRA 78.3 %), HIIT (SUCRA 54.8 %), AE (SUCRA 34.8 %), CON (SUCRA 29.3 %), and ME (SUCRA 3.3 %).

3.5. Risk of bias and publication bias

Concerning bias and publication bias, our analysis scrutinized several factors. With respect to risk of bias, 28 studies addressed issues of insufficient randomization, and 26 studies provided complete outcome data. However, adequacy of allocation concealment was present in only 8 studies. Additionally, while blinding was practiced in 14 studies, it was absent in 4, and proper blinding of outcome assessment was observed in merely 12 studies. On the positive side, 29 studies showed good reporting selectivity.

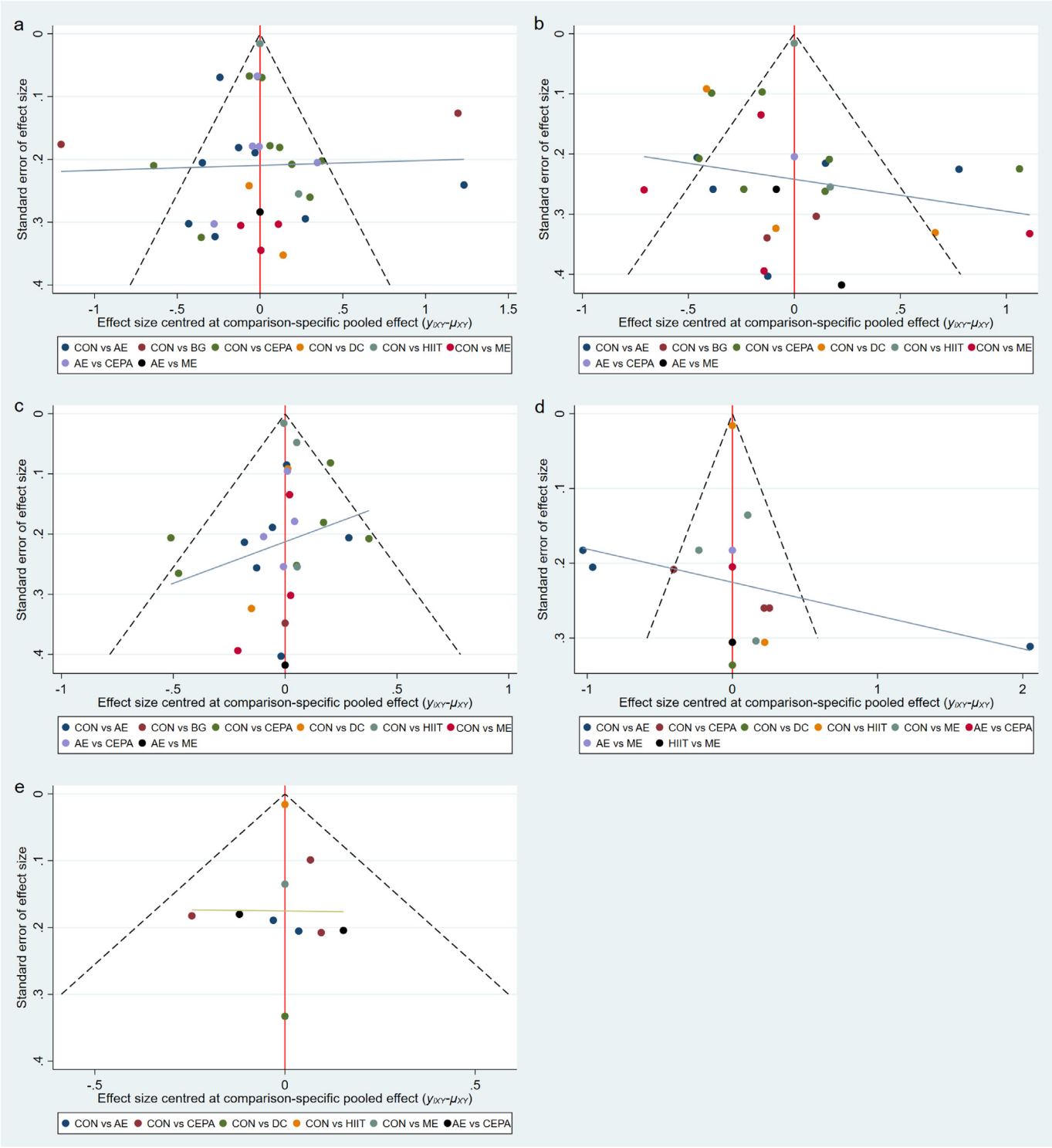

Fig. 4. The Funnel plot for a: updating accuracy, b: inhibition accuracy, c: inhibition reaction time, d: shifting accuracy, e: shifting reaction time.

As for publication bias, we utilized funnel plots (Fig. 4a, b, c, d and e) to identify its potential presence. The scatter points in the plots were found to be symmetrically distributed on either side of the vertical axis, suggesting varying levels of publication bias. Specifically, Fig. 4c and e demonstrated evenly distributed points, while Fig. 4a and b indicated some presence of bias, and Fig. 4d showed notable bias. Despite these individual instances, our overall analysis did not detect significant publication bias across all included studies.

4. Discussion

This is the first systematic review and network meta-analysis that examined the impact of various PA types on three core components of executive function—updating, inhibition, and shifting—several key findings have emerged to substantiate the cognitive benefits of PA. Our analysis corroborates the hypothesis that targeted PA can significantly improve specific EFs. Specifically, (1) BG outperformed all other PA types in enhancing the accuracy of updating, securing a SUCRA score of 94.4 %; (2) CEPA was the most effective in improving inhibition accuracy, as demonstrated by its leading SUCRA score of 71.7 %, followed closely by ME and DC; (3) For reaction time in inhibition, BG had the highest SUCRA score of 94.8 %, suggesting its efficacy; (4) While no single PA modality displayed a statistically significant advantage in ameliorating shifting accuracy, AE did procure the highest SUCRA score, registering at 77.5 %; (5) DC manifested a remarkable reduction in reaction time in shifting tasks, evidenced by its exceptional SUCRA score of 99.5 %.

Our results showed that BG in enhancing updating accuracy and reducing inhibition reaction time in children and adolescents. The meta-analysis showed that BG is particularly effective in improving sub-functions of these EFs, which corroborates existing literature.[36,37] BG involve complex maneuvers, such as dribbling past opponents, which require the integration of visuospatial processing to execute complex "automatic" actions.[38] Optimal performance in these activities requires intricate working memory operations to continuously update and maintain various types of information,[39] such as player positions and tactical decisions. The interplay between motor and cognitive systems can be traced back to common neural mechanisms, particularly those in the cerebellum.[40,41] Furthermore, rapid action decisions, such as passing or catching the ball, are required in BG; this rapid decision-making process aids in enhancing the speed of inhibition. And, sustained attention is required to adapt to the ever-changing dynamics of the sport and game environment, which helps to enhance the ability to resist external distractions and, consequently improve inhibition reaction time.[42,43]

Interestingly, while BG excels in reducing inhibition reaction time, their efficacy in enhancing inhibition accuracy is moderate. We found that CEPA was the most effective intervention for improving this dimension of inhibition. We propose that the discrepancy in the efficacy of different types of PA in improving two facets of inhibition—accuracy and reaction time—can be attributed to their respective demands and the inherent characteristics of the PAs. Inhibition reaction time benefits from quick decision-making and responses,[38,44] aligning well with the demands of BG. Inhibition accuracy, on the other hand, requires meticulousness in tasks that require precision and focus on the correctness of actions. CEPA often involves complex, higher-order cognitive engagement and processing, requiring participants to think more deeply about the accuracy of their decisions and actions more deeply, thereby enhancing inhibition accuracy.[22,23]

Additionally, our findings indicate that CEPA also yields beneficial effects on inhibition reaction time, shifting accuracy, and shifting reaction time. This is consistent with prior research; for instance, a meta-analysis by Song et al. revealed that although the effect size brought about by CEPA was relatively small (SMD = 0.21), it remains an effective means of improving children's EFs, particularly inhibition and shifting.[45] These benefits primarily stem from the high cognitive demands of CEPA,[46] which enhance neural plasticity and strengthen connections in brain regions associated with motor and cognitive functions,[47,48] eliciting a notable synergistic effect.[49,50] Specifically, peak neural activation occurs when exercise tasks are difficult, novel, attention-intensive, and involve unpredictable responses. This implies that CEPA furnishes the necessary neural stimulation to co-activate neurons across various brain regions, thereby facilitating the enhancement of EFs.[51] Moreover, the complexity of CEPA necessitates deeper cognitive and social engagement from children, which activates neural networks closely associated with EFs,[52,53] particularly in the prefrontal cortex.[54] Hence, given the broad beneficial impacts of CEPA on multiple core sub-components of EFs in children and adolescents, it can be considered a primary choice for PA interventions.

Our study adds important data on the value of DC as an effective PA for enhancing updating and shifting in children and adolescents. Specifically, DC ranked second in improving updating accuracy and shifting accuracy, while it took the top position in reducing shifting reaction time. Even more, DC was significantly better at reducing shifting reaction time compared to other types of PA. This supports a wealth of existing research indicating that DC is a powerful tool for nurturing cognitive flexibility in young individuals.[55,56] During DC, participants follow the music's tempo, continuously adapting their dance moves. This synchronization between movement and music enriches cognitive adaptability, making the children more mentally agile.[55] Additionally, the rhythmic elements inherent in the music used in DC training serve as a significant catalyst for the development of cognitive flexibility in children and adolescents. Our findings are corroborated by other studies that have demonstrated the beneficial impacts of short-term, six-month music training sessions on children's behavior and synaptic growth.[57] Such outcomes suggest that musical training effectively stimulates activities in the cortical regions of the brain, promoting brain plasticity.[57] Comparable to the benefits observed in BG, DC's positive impact on updating can be traced back to its intrinsic requirements. Participants are called upon to memorize an array of details, such as the names, features, and spatial arrangements of particular dance moves. This cognitive exercise involves ongoing sensorimotor stimuli that engage participants in a wide variety of bodily movements. They are required to remember and recall extensive sequences of actions, storing this information for later use.[58,59] This enriching cognitive process taps into the brain's prefrontal regions associated with working memory,[60,61] consequently fostering their ability to update working memory efficiently.[55] Therefore, the complex choreographic and rhythmic elements in dance not only contribute to better cognitive flexibility but also stimulate brain areas crucial for memory and inhibitory control. Given these comprehensive benefits, dance should be considered leading candidates for interventions aimed at augmenting updating and shifting of EFs in children and adolescents.

In the studies we included, AE emerged as a predominant intervention type for EFs. Although AE achieved the highest rank in terms of shifting accuracy, it showed moderate efficacy in other evaluation metrics. While extant literature posits that AE augments cerebral blood flow and oxygen content,[62] thereby stimulating hippocampal neurogenesis[63] and enhancing cognitive efficiency,[64] its impact on EFs appears limited. This may be attributed to the inherently cardiovascular focus of AE, which necessitates a lower cognitive load compared to DC and BG. Consequently, AE may elicit less activation in brain regions associated with EFs. Conversely, DC and BG require simultaneous processing of multiple informational streams and tasks, which engages a broader array of cognitive domains, including the prefrontal cortex, parietal lobe, and cerebellum.[51] This multi-faceted engagement arguably fosters enhanced cognitive integration and coordination, contributing to improved EFs.[54] Moreover, AE activities like running and skipping often lack the diverse, game-like elements found in BG and CEPA, which arguably make the latter more engaging. This facet is particularly crucial for long-term adherence to PA regimes, which is essential for achieving optimal EFs outcomes.[27] Thus, the lower PA adherence observed in certain types of PA, such as HIIT and AE, may partly explain their suboptimal

effectiveness in EFs interventions.[65] However, our analysis is constrained by the limited number of studies reporting participant compliance—only eight of the included studies offered such data. This precludes a more comprehensive examination of compliance as a moderating variable on intervention outcomes. Future research could provide valuable insights by thoroughly analyzing the effects of varying compliance with different types of PA on EFs in children and adolescents.

Building on our discussion, it's noteworthy that in this study, both BG and DC registered SMDs > 1 in enhancing the updating of EFs. This differs significantly from the existing literature on PA assessments targeting EFs in children and adolescents, which typically report only modest effects. We propose that this discrepancy is due to the distinct advantages that BG and DC offer in improving updating accuracy. Unlike previous studies, which often combine these specific activities with other types of physical exercise, our research isolates the effects of BG and DC, thereby providing a more nuanced understanding of their effectiveness. This approach is likely to accentuate the observed effect sizes for these activities, explaining why the SMDs in our study are higher than those in previous research. This finding further highlights the unique benefits that BG and DC bring to the development of EFs in young populations, particularly updating.

Although this NMA was based on high quality RCTs, there are several limitations to consider. First, the sample population was restricted to healthy children and adolescents, thereby excluding overweight and obese populations. Evidence suggests that BMI may modulate the long-term influence of PA on EFs,[66,67] potentially offering greater benefits to children with elevated BMI.[3] Second, the intervention protocols, which differ in cycle length, frequency, and intensity, represent a critical variable that we did not examine in subgroup analyses. Future investigations could aim to identify the optimal intervention 'dose' for each PA type to maximize its impact on EFs in young populations. Third, the study did not encompass resistance training, a significant form of PA, due to the absence of research meeting the inclusion criteria. Thus, the potential benefits of resistance training on EFs remain inadequately explored. Fourth, the scarcity of studies reporting participant compliance introduces a potential bias and heterogeneity in our findings.

5. Conclusion

In conclusion, this NMA confirms that PA has clear benefits on EFs in children and adolescents and also shows that the magnitude of the effect varies according to the type of PA. Our results indicate that BG was most effective in improving updating accuracy and inhibition reaction time, while CEPA was most effective in improving inhibition accuracy. DC is advantageous for updating accuracy and shifting reaction time. AE, although prevalent in the literature, showed moderate efficacy across EFs components. These findings provide a strong basis for future research aimed at designing targeted PA interventions to maximize improvements in EFs in young populations, ultimately promoting their holistic growth and development.

Funding information

This work was supported by grants from the General Project of the National Social Science Foundation of China (No. 18BTY095 [to J. Z]).

Confirmation of ethical compliance

No ethical approval and consent was required for the current study.

CRediT authorship contribution statement

Junyu Wang: Methodology, Writing – original draft. **Yong Yang:** Writing – review & editing. **Longkai Li:** Investigation, Validation. **Xuemin Yang:** Resources, Formal analysis. **Xuejun Guo:** Resources, Formal analysis. **Xun Yuan:** Software, Visualization. **Ting Xie:** Data curation, Validation. **Kunyi Yang:** Software, Formal analysis. **Jie Zhuang:** Conceptualization, Funding acquisition.

Declaration of interest statement

Authors declare no conflict of interest.

Acknowledgment

Not applicable.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jsams.2023.11.006.

References

1. Diamond A. Executive functions. *Annu Rev Psychol* 2013;64:135-168.
2. Spiegel JA, Goodrich JM, Morris BM et al. Relations between executive functions and academic outcomes in elementary school children: a meta-analysis. *Psychol Bull* 2021;147(4):329-351.
3. Xue Y, Yang Y, Huang T. Effects of chronic exercise interventions on executive function among children and adolescents: a systematic review with meta-analysis. *Br J Sports Med* 2019;53(22):1397-1404.
4. Zeng X, Cai L, Wong SH et al. Association of sedentary time and physical activity with executive function among children. *Acad Pediatr* 2021;21(1):63-69.
5. Jirout J, LoCasale-Crouch J, Turnbull K et al. How lifestyle factors affect cognitive and executive function and the ability to learn in children. *Nutrients* 2019;11(8).
6. Fan X, Cao ZB. Physical activity among Chinese school-aged children: national prevalence estimates from the 2016 physical activity and fitness in China-the youth study. *J Sport Health Sci* 2017;6(4):388-394.
7. Schoemaker K, Bunte T, Wiebe SA et al. Executive function deficits in preschool children with ADHD and DBD. *J Child Psychol Psychiatry Allied Discip* 2012;53(2):111-119.
8. Hai T, Duffy HA, Lemay JA et al. Impact of stimulant medication on behaviour and executive functions in children with attention-deficit/hyperactivity disorder. *World J Clin Pediatr* 2022;11(1):48-60.
9. Miklós M, Futó J, Komáromy D et al. Executive function and attention performance in children with ADHD: effects of medication and comparison with typically developing children. *Int J Environ Res Public Health* 2019;16(20).
10. Ameis SH, Blumberger DM, Croarkin PE et al. Treatment of executive function deficits in autism spectrum disorder with repetitive transcranial magnetic stimulation: a double-blind, sham-controlled, pilot trial. *Brain Stimul* 2020;13(3):539-547.
11. Rosenau PT, Openneer TJC, Matthijssen AM et al. Effects of methylphenidate on executive functioning in children and adolescents with ADHD after long-term use: a randomized, placebo-controlled discontinuation study. *J Child Psychol Psychiatry Allied Discip* 2021;62(12):1444-1452.
12. Spencer-Smith M, Quach J, Mensah F et al. The effectiveness of working memory training for children with low working memory. *Pediatrics* 2020;146(6).
13. Wu CS, Shang CY, Lin HY et al. Differential treatment effects of methylphenidate and atomoxetine on executive functions in children with attention-deficit/hyperactivity disorder. *J Child Adolesc Psychopharmacol* 2021;31(3):187-196.
14. Xue Y, Yang Y, Huang T. Effects of chronic exercise interventions on executive function among children and adolescents: a systematic review with meta-analysis. *Br J Sports Med* 2019;53(22):1397-1404.
15. Martín-García M, Alegre LM, García-Cuartero B et al. Effects of a 3-month vigorous physical activity intervention on eating behaviors and body composition in overweight and obese boys and girls. *J Sport Health Sci* 2019;8(2):170-176.
16. MacKelvie KJ, Khan KM, Petit MA et al. A school-based exercise intervention elicits substantial bone health benefits: a 2-year randomized controlled trial in girls. *Pediatrics* 2003;112(6 Pt 1):e447.
17. Liang X, Qiu H, Wang P et al. The impacts of a combined exercise on executive function in children with ADHD: a randomized controlled trial. *Scand J Med Sci Sports* 2022;32(8):1297-1312.
18. Wu Q, Niu X, Zhang Y et al. A comparative study of inhibition function between high-intensity interval training and moderate-intensity continuous training in healthy people: a systematic review with meta-analysis. *Int J Environ Res Public Health* 2023;20(4).
19. Amatriain-Fernández S, Ezquerro García-Noblejas M, Budde H. Effects of chronic exercise on the inhibitory control of children and adolescents: a systematic review and meta-analysis. *Scand J Med Sci Sports* 2021;31(6):1196-1208.
20. Zhang M, Garnier H, Qian G et al. Effect of 11 weeks of physical exercise on physical fitness and executive functions in children. *Children (Basel, Switzerland)* 2023;10(3).
21. Koutsandréou F, Wegner M, Niemann C et al. Effects of motor versus cardiovascular exercise training on children's working memory. *Med Sci Sports Exerc* 2016;48(6):1144-1152.
22. Vazou S, Pesce C, Lakes K et al. More than one road leads to Rome: a narrative review and meta-analysis of physical activity intervention effects on cognition in youth. *Int J Sport Exerc Psychol* 2019;17(2):153-178.

23. Wang CH, Moreau D, Yang CT et al. Aerobic exercise modulates transfer and brain signal complexity following cognitive training. *Biol Psychol* 2019;144:85-98.
24. Dias S, Ades AE, Welton NJ et al. *Network Meta-analysis for Decision Making*, 2018.
25. Leucht S, Chaimani A, Cipriani AS et al. Network meta-analyses should be the highest level of evidence in treatment guidelines. *Eur Arch Psychiatry Clin Neurosci* 2016;266(6):477-480.
26. Ludyga S, Held S, Rappelt L et al. A network meta-analysis comparing the effects of exercise and cognitive training on executive function in young and middle-aged adults. *Eur J Sport Sci* 2022;1-11.
27. Zhu F, Zhu X, Bi X et al. Comparative effectiveness of various physical exercise interventions on executive functions and related symptoms in children and adolescents with attention deficit hyperactivity disorder: a systematic review and network meta-analysis. *Front Public Health* 2023;11:1133727.
28. Page MJ, McKenzie JE, Bossuyt PM et al. Updating guidance for reporting systematic reviews: development of the PRISMA 2020 statement. *J Clin Epidemiol* 2021;134:103-112.
29. Kenney W. *Physiology of Sport and Exercise W/Web Study Guide, 5th edition.*, Human Kinetics, 2012.
30. Lipszyc J, Schachar R. Inhibitory control and psychopathology: a meta-analysis of studies using the stop signal task. *J Int Neuropsychol Soc* 2010;16(6):1064-1076.
31. Miyake A, Friedman NP. The nature and organization of individual differences in executive functions: four general conclusions. *Curr Dir Psychol Sci* 2012;21(1):8-14.
32. Drevon D, Fursa SR, Malcolm AL. Intercoder reliability and validity of WebPlotDigitizer in extracting graphed data. *Behav Modif* 2017;41(2):323-339.
33. Higgins J, Green SR. *Cochrane Handbook for Systematic Review of Interventionsversion 5.1.0*, 2011.
34. Li L, Zhang J, Cao M et al. The effects of chronic physical activity interventions on executive functions in children aged 3-7 years: a meta-analysis. *J Sci Med Sport* 2020;23(10):949-954.
35. DerSimonian R, Laird N. Meta-analysis in clinical trials. *Control Clin Trials* 1986;7(3):177-188.
36. Law CK, Lam FM, Chung RC et al. Physical exercise attenuates cognitive decline and reduces behavioural problems in people with mild cognitive impairment and dementia: a systematic review. *J Physiother* 2020;66(1):9-18.
37. Wang JG, Cai KL, Liu ZM et al. Effects of mini-basketball training program on executive functions and core symptoms among preschool children with autism spectrum disorders. *Brain Sci* 2020;10(5).
38. Policastro F, Accardo A, Marcovich R et al. Relation between motor and cognitive skills in Italian basketball players aged between 7 and 10 years old. *Sports (Basel)* 2018;6(3).
39. Newell KM. Coordination, control and skill. *Adv Psychol* 1985;27:295-317.
40. Rigoli D, Piek JP, Kane R et al. An examination of the relationship between motor coordination and executive functions in adolescents. *Dev Med Child Neurol* 2012;54(11):1025-1031.
41. Diamond A. Close interrelation of motor development and cognitive development and of the cerebellum and prefrontal cortex. *Child Dev* 2000;71(1):44-56.
42. Chen CH, Rekik G, Belkhir Y et al. Gender differences in attention adaptation after an 8-week FIFA 11(+) for kids training program in elementary school children. *Children (Basel, Switzerland)* 2021;8(9).
43. de Greeff JW, Bosker RJ, Oosterlaan J et al. Effects of physical activity on executive functions, attention and academic performance in preadolescent children: a meta-analysis. *J Sci Med Sport* 2018;21(5):501-507.
44. Zoudji B, Thon B, BJPoS D. Exercise. Efficiency of the mnemonic system of expert soccer players under overload of the working memory in a simulated decision-making task. 2010;11(1):18-26.
45. Song W, Feng L, Wang J et al. Play smart, be smart? Effect of cognitively engaging physical activity interventions on executive function among children 4~12 years old: a systematic review and meta-analysis. *Brain Sci* 2022;12(6).
46. Pesce C. Shifting the focus from quantitative to qualitative exercise characteristics in exercise and cognition research. *J Sport Exerc Psychol* 2012;34(6):766-786.
47. Best JR. Effects of physical activity on children's executive function: contributions of experimental research on aerobic exercise. *Dev Rev* 2010;30(4):331-551.
48. Tomporowski, Phillip D, Mccullick BA et al. *Enhancing Children's Cognition With Physical Activity Games*, 2015.
49. Bedard C, Bremer E, Graham JD et al. Examining the effects of acute cognitively engaging physical activity on cognition in children. *Front Psychol* 2021;12:653133.
50. Kassai R, Futo J, Demetrovics Z et al. A meta-analysis of the experimental evidence on the near- and far-transfer effects among children's executive function skills. *Psychol Bull* 2019;145(2):165-188.
51. Diamond A, Ling DS. Conclusions about interventions, programs, and approaches for improving executive functions that appear justified and those that, despite much hype, do not. *Dev Cogn Neurosci* 2016;18:34-48.
52. Mazzoli E, Koorts H, Salmon J et al. Feasibility of breaking up sitting time in mainstream and special schools with a cognitively challenging motor task. *J Sport Health Sci* 2019;8(2):137-148.
53. Schmidt M, Benzing V, Kamer M. Classroom-based physical activity breaks and children's attention: cognitive engagement works! *Front Psychol* 2016;7:1474.
54. Mazzoli E, Salmon J, Teo WP et al. Breaking up classroom sitting time with cognitively engaging physical activity: behavioural and brain responses. *PloS One* 2021;16(7):e0253733.
55. Zinelabidine K, Elghoul Y, Jouira G et al. The effect of an 8-week aerobic dance program on executive function in children. *Percept Mot Skills* 2022;129(1):153-175.
56. Shen Y, Zhao Q, Huang Y et al. Promotion of street-dance training on the executive function in preschool children. *Front Psychol* 2020;11:585598.
57. Moreno S, Marques C, Santos A et al. Musical training influences linguistic abilities in 8-year-old children: more evidence for brain plasticity. *Cereb cortex (New York, NY: 1991)* 2009;19(3):712-723.
58. Merom D, Cumming R, Mathieu E et al. Can social dancing prevent falls in older adults? A protocol of the Dance, Aging, Cognition, Economics (DAnCE) fall prevention randomised controlled trial. *BMC Public Health* 2013;13:477.
59. Jola C, McAleer P, Grosbras MH et al. Uni- and multisensory brain areas are synchronised across spectators when watching unedited dance recordings. *i-Perception* 2013;4(4):265-284.
60. van Ewijk H, Weeda WD, Heslenfeld DJ et al. Neural correlates of visuospatial working memory in attention-deficit/hyperactivity disorder and healthy controls. *Psychiatry Res* 2015;233(2):233-242.
61. Klingberg T, Forssberg H, Westerberg H. Increased brain activity in frontal and parietal cortex underlies the development of visuospatial working memory capacity during childhood. *J Cogn Neurosci* 2002;14(1):1-10.
62. Zauner A, Doppenberg E, Woodward JJ et al. Multiparametric continuous monitoring of brain metabolism and substrate delivery in neurosurgical patients. *Neurol Res* 1997;19(3):265-273.
63. Churchill JD, Galvez R, Colcombe S et al. Exercise, experience and the aging brain. *Neurobiol Aging* 2002;23(5):941-955.
64. Byun K, Hyodo K, Suwabe K et al. Positive effect of acute mild exercise on executive function via arousal-related prefrontal activations: an fNIRS study. *Neuroimage* 2014;98:336-345.
65. Krafft CE, Schaeffer DJ, Schwarz NF et al. Improved frontoparietal white matter integrity in overweight children is associated with attendance at an after-school exercise program. *Dev Neurosci* 2014;36(1):1-9.
66. Huang T, Tarp J, Domazet SL et al. Associations of adiposity and aerobic fitness with executive function and math performance in Danish adolescents. *J Pediatr* 2015;167(4):810-815.
67. Raine LB, Khan NA, Drollette ES et al. Obesity, visceral adipose tissue, and cognitive function in childhood. *J Pediatr* 2017;187:134-140.e13SS3.